

Trace of Relevant Analysis Differences

Science 2024 public version / v1 compared with the current Oxygen Dynamics pipeline. Prepared for internal interpretation and README support. Generated 2026-06-01.

Plain-Language Summary

The current pipeline is not just a cleaned-up copy of the Science 2024 analysis. Most detection parameters are intentionally preserved, but several important parts are now safer, more explicit, and more event-based. The most important interpretation point is that the current sink amplitude metric is raw-signal based, whereas the old Science/v1 value called NormOxySinkAmp was calculated from the processed detection trace.

Therefore, lower current amplitudes do not automatically mean the biology changed or that the detector is failing. They mostly reflect that the current output reports a different and more physically interpretable amplitude definition.

Highest-Impact Differences

Topic	Science 2024 / v1	Current pipeline	Likely impact
Sink amplitude	NormOxySinkAmp was calculated from the processed/convolved detection trace relative to a fitted trend.	NormOxySinkAmp / NormOxySinkAmpPercent is calculated from the original raw TIFF trace as raw baseline minus raw event minimum, normalized to baseline. DetectionOxySinkAmp preserves the detection-domain value.	Current amplitude is usually lower and should not be directly compared to the old Science NormOxySinkAmp. For legacy comparison, inspect DetectionOxySinkAmp.
Denoised TIFF use	Science public version had basic DENOISED filename handling, but TIFF selection was loose and could silently overwrite raw data if multiple TIFF files were present.	Exactly one raw TIFF and at most one denoised TIFF are identified. Denoised data are used for detection, raw data for quantification, with dimension checks and AnalysisInfo provenance.	More reproducible. Denoised detection can increase event counts, while raw quantification may reduce average amplitude.
Pixel size	Area metrics depended on the old pixel size setting.	Corrected pixel size, e.g. 2.35 to 6.75 um/pixel.	Physical area changes by about 8.25-fold for that example. Area, burden, and 1 mm2 normalization change strongly. Pixel-based detection thresholds do not automatically change.
Event-level outputs	Stats were mainly ROI/pocket summarized. Individual-event analysis was partly separate.	Event-level tables and burden outputs are integrated into stats.	Better for biological event analysis, but not always directly comparable to older ROI-level tables.

1. Amplitude Definition

This is the main reason why the current sink amplitudes can look smaller than in the original Science analysis.

In the Science/v1 scripts, NormOxySinkAmp was calculated from Mean_OxySink_Trace_Convo and a polynomial trend. This means it was a processed detection-domain amplitude, not a direct raw percent signal change.

In the current pipeline, NormOxySinkAmp and NormOxySinkAmpPercent are calculated from the original raw TIFF trace. The calculation is baseline minus event minimum, divided by the raw baseline. This is more interpretable as a raw percent signal drop.

For direct comparison to the historical Science-style metric, DetectionOxySinkAmp is the closer current output.

2. Denoised Detection and Raw Quantification

The current version separates detection and quantification more clearly. The denoised image can be used to detect oxygen sinks, but the amplitude and raw trace quantification are taken from the original TIFF.

This can produce more detected events than the older analysis, especially if denoising improves contrast or continuity. At the same time, the raw signal amplitude of each event can be smaller than the processed detection-domain amplitude.

The output now records the raw TIFF, denoised TIFF, detection source, and quantification source so this can be audited later.

3. Pixel Size and Area-Dependent Metrics

Changing the pixel size from 2.35 to 6.75 um/pixel changes physical area by $(6.75 / 2.35)^2$, which is approximately 8.25.

Therefore, OxySinkArea_um, event area, recording area, normalized sinks per 1 mm², and hypoxic burden values are strongly affected.

The core detection area thresholds remain pixel-based. So the same object in pixels can still pass the detector, but its reported physical size changes substantially.

4. Detection, Tracking, and Refinement

Most core detection settings were preserved from the Science script: smoothing, percentile threshold, minimum and maximum sink size, circularity threshold, event duration thresholds, and trace-correlation refinement are intentionally similar.

The current version moves this logic into helper functions. This makes the code easier to audit and maintain, but the intended behavior remains aligned with the original analysis.

Some edge cases can differ because the current pipeline uses safer fallback correlation and z-scoring helpers when MATLAB toolbox functions are unavailable.

5. Stats and Output Structure

The current stats pipeline is more explicit about recording identity, repeated mouse IDs, condition handling, output provenance, and event-level metrics.

It also uses safer mean calculations for empty or NaN-containing event cells. This can create small differences compared with direct mean calculations in the older scripts.

The newer event-level and burden sheets are better suited for current biological comparisons, but they should be interpreted as an expanded analysis rather than a pure reproduction of the old Science tables.

Practical Interpretation

- Do not directly compare current NormOxySinkAmpPercent with the Science paper NormOxySinkAmp. They are different amplitude definitions.
- Use current NormOxySinkAmpPercent for raw biological signal interpretation.
- Use DetectionOxySinkAmp when asking how the current data compare with the older processed detection-domain amplitude.
- When comparing counts or burden across versions, always document whether denoised detection was used and which pixel size was applied.
- For publications or internal reports, state that the updated pipeline uses denoised data for detection where available, but raw data for amplitude quantification.

Bottom Line

The current pipeline is more reproducible and more biologically interpretable for raw signal quantification, but it is not numerically identical to the original Science 2024 analysis. The largest differences are the raw-based amplitude definition, denoised-assisted detection, corrected pixel size, and integrated event-level statistics.